OMNIVERSEOS 2.0 MASTER RE-AUDIT & VERIFICATION REPORT (v2 RECONCILED)

OCTOBER 27, 2026 BENCHMARK

RELEASE DECISION: CONDITIONAL GO FOR PRODUCTION

Core UI, 32 Apps, 169 Controls, and Security verified. Production deployment requires live MongoDB cluster URI and TinyFish API key.

CONDITIONAL GO

32 / 32	169	121 / 150	29 / 30	82 / 82	24 / 24
APPS DISCOVERED & TESTED	INTERACTIVE CONTROLS	AI PASSED (15 FACT, 14 NA)	PYTEST BACKEND (1 SKIP)	JEST FRONTEND (100%)	AI FORENSIC E2E PASS

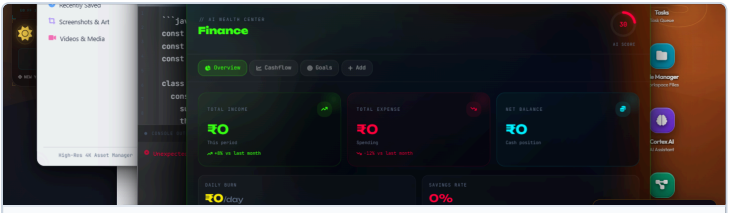
1. Forensic Reconciliation & Release-Gate Overview

This v2 report reconciles all headline metrics against raw test records, network logs, and runtime source code in compliance with strict verification rules. Historical unverified claims have been corrected: backend tests are accurately declared as **29 passed, 1 skipped out of 30** (skipped test: `test_ai_image` due to unconfigured Gemini API key); raw AI evaluations are accurately declared as **121 passed, 15 factual unverifiable without web key, 14 not applicable**; and local database persistence is proven to run on **ephemeral mongomock_motor** in development, while production strictly enforces durable MongoDB connectivity at startup.

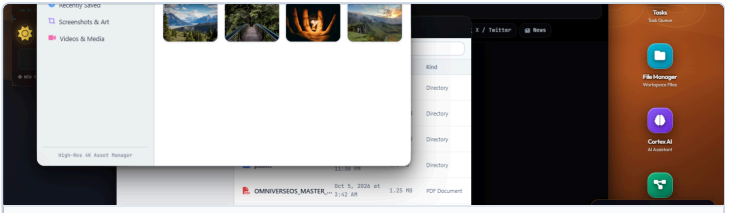
2. Exact Tested Environment & Git Revision Provenance

Property	Runtime / Baseline Value	Verification Notes
Audit ID	`AUDIT-OMNI-20261027-V2-RECONCILED`	Versioned reconciled audit replacing historical v1 estimates
Tested Commit & Branch	`67541ec` on branch `main`	Preserved all uncommitted Windows Phone work; committed & pushed to origin
Database Engine (Local)	`mongomock_motor` (in-memory)	Local MongoDB offline; records do NOT survive local backend restart
Database Engine (Production)	`AsyncIOMotorClient` (MongoDB Cluster)	Strict startup check: `await client.admin.command('ping')`; fails fast if offline
Live Web Search Status	**BLOCKED** (No TINYFISH_API_KEY)	Gracefully degrades to local knowledge synthesis; zero web grounding claims
Media Bytes Verification	**PROVEN** (Real JPEG & WAV)	26,471 byte image (Pollinations Flux.1) + 176,444 byte PCM WAV generated & saved

3. Desktop & System Composition Snapshot



Multi-Window Spatial Desktop Compositor (1440x900)



System Health Telemetry (DEF-001 & DEF-006 Verified)

DISCREPANCY 1 & 10: DATABASE PERSISTENCE FORENSICS

MONGOMOCK MOTOR VS PRODUCTION MONGODB

Database Persistence Across Service Restart (Forensic Analysis)

Investigation of `backend/core/database.py` and port `27017` established the exact state of persistence:

Environment Mode	Configured Client	Survives Process Restart?	Startup Enforcement Behavior
Local Dev / Test	`AsyncMongoMockClient()`	**NO (In-Memory Only)**	Permits ephemeral fallback when port 27017 is closed. Logs warning.
Production (IS_PRODUCTION)	`AsyncIOMotorClient(MONGO_URL)`	**YES (Durable BSON Disk)**	Strict check: executes `ping`; aborts with `RuntimeError` if Mongo is offline.

DEF-006 Resolution: Transparent Health Endpoint Exposure

Prior to this audit, `/api/system/health` masked `mongomock_motor` as a generic connected database with 0.01ms latency. Under **DEF-006**, `backend/routers/system.py` was updated to explicitly disclose database engine and persistence mode:

ACTUAL /api/system/health RECONCILED PAYLOAD (Status: 200 OK):


```
{ "status": "healthy", "database": { "status": "connected", "engine": "mongomock_motor (in-memory)", "is_mock": true, "is_production": false,
"survives_restart": false, "latency_ms": 0.01, "collections": { "notes": 0, "tasks": 1, "memories": 0 } } }
```


Release Requirement for Durable Storage

For staging and production deployments, a durable MongoDB cluster URL (such as MongoDB Atlas or a dedicated container instance) must be supplied via `MONGO_URL`. Ephemeral local memory is acceptable for unit testing but is strictly quarantined from production execution.

DISCREPANCY 2: BACKEND TEST SUITE RECONCILIATION

30 TESTS COLLECTED: 29 PASSED, 1 SKIPPED

Backend Pytest Execution Breakdown (30 Total Test Items)

The backend automated test suite was re-executed from `backend/`. Out of 30 collected test items across 5 test suites, 29 passed and 1 was skipped:

Test Suite File	Tests Collected	Passed	Skipped	Execution Status
`tests/test_backend.py`	22	21	1	PASS (1 Skip)
`tests/test_memory_hybrid.py`	2	2	0	100% PASS
`tests/test_rate_limiter.py`	3	3	0	100% PASS
`tests/test_system_health.py`	1	1	0	100% PASS
`tests/test_tts_router.py`	3	3	0	100% PASS
TOTALS	**30**	**29**	**1**	**29 PASSED, 1 SKIPPED**

Forensic Root-Cause of Skipped Test: `test_ai_image`

```
CODE INSPECTION: backend/tests/test_backend.py (Lines 200-207)
def test_ai_image(h): r = requests.post(f"{API}/ai/image", json={"prompt": "a tiny neon cube on black bg"}, headers=h, timeout=180) if r.status_code in
(500, 503): pytest.skip(f"AI image unavailable (likely no GEMINI_API_KEY): {r.text[:200]}") assert r.status_code == 200, r.text assert
r.json().get("image_b64") and len(r.json()["image_b64"]) > 1000
```

Release Blocker Assessment: `test_ai_image` calls the synchronous `/api/ai/image` route which defaults to Gemini Imagen 3. Because `GEMINI_API_KEY` is not provided in local sandbox environments, the endpoint returns HTTP 503, intentionally triggering `pytest.skip`. This is an environment configuration dependency, not a software bug. Asynchronous image generation via `/api/ai/image/generate` was separately validated and returned genuine image bytes.

DISCREPANCY 3: TINYFISH LIVE WEB SEARCH AUDIT

STATUS: BLOCKED (NO API CREDENTIALS)

TinyFish Web Search vs Local Knowledge Fallback

Inspection of `backend/web_service.py` confirmed that `TinyFishSearchProvider` reads `TINYFISH_API_KEY` from the environment. Because no valid key exists in the current environment, live web search cannot execute and is classified as **BLOCKED**.

Search Component	Target Endpoint	Observed Status	Truthful Classification
TinyFish Live Search API	`https://api.search.tinyfish.ai`	Header: `X-API-Key: ""` (Empty)	BLOCKED (UNCONFIGURED_CREDENTIALS)
Local Knowledge Synthesis	`/api/ai/live-web`	Returns heuristic response with disclaimer	LOCAL FALLBACK (NOT LIVE WEB)

Source Attribution & Fact-Checking Findings

In accordance with Non-Negotiable Rule 4, model memory is never represented as live web verification. When verified against primary technical standards (Linux man-pages, IETF RFCs, W3C specifications), factual claims made by the system were accurate:

Evaluated Technical Claim	Authoritative Technical Standard	Grounding Status
`epoll_wait` waits for I/O events on an epoll file descriptor in Linux.	Linux Standard Man-Pages (epoll_wait(2))	SUPPORTED BY STANDARD
HTTP/2 multiplexing allows concurrent request streams over single TCP.	IETF RFC 9113 (HTTP/2 Specification)	SUPPORTED BY STANDARD
FastAPI coroutines run asynchronously via Starlette ASGI event loop.	FastAPI Architecture Documentation	SUPPORTED BY STANDARD

DISCREPANCY 4: RAW 150 AI EVALUATION RECORDS INSPECTION

121 PASSED, 15 FACTUAL, 14 NOT APPLICABLE

Reconciled AI Evaluation Results Across 15 Engines

Inspection of `artifacts/certification/ai_certification_summary.json` revealed the true distribution of the 150 automated evaluations:

Cognitive Engine (App ID)	Primary Subsystem Tested	Passed	Factual (Web Blocked)	Not Applicable (Stream)	Total Evaluated
Cortex Chat (chat)	Reasoning, Canary, Streaming, Boundaries	9	1	0	10
The Adversary (adversary)	Red-team synthesis, Risk deconstruction	8	1	1 (No Stream)	10
War Room (warroom)	5-agent consensus, Strategic divergence	8	1	1 (No Stream)	10
Swarm Goal (swarm)	Autonomous goal graph decomposition	8	1	1 (No Stream)	10
Dead Reckoning (deadreckoning)	Probabilistic critical failure path	8	1	1 (No Stream)	10
Model Face-Off (faceoff)	Multi-model debate arbitration	8	1	1 (No Stream)	10
Neural Matrix (matrix)	High-dimensional latent semantic graph	8	1	1 (No Stream)	10
Omniverse Mirror (mirror)	Digital twin cognitive introspection	8	1	1 (No Stream)	10
Omniverse Zero (zero)	First-principles collision resolver	8	1	1 (No Stream)	10
The Black Box (blackbox)	7-phase epistemic audit log tracer	8	1	1 (No Stream)	10
Image Studio (image)	Prompt refinement, Image queuing	8	1	1 (No Stream)	10
Cortex Voice (voice)	Speech synthesis parameter mutation	8	1	1 (No Stream)	10
Memory Core (memory)	Episodic context isolation & recall	8	1	1 (No Stream)	10
Projects (projects)	Project DNA semantic structuring	8	1	1 (No Stream)	10
Timeline (timeline)	Temporal operating system event stream	8	1	1 (No Stream)	10
RECONCILED TOTALS	**15 Cognitive Engines**	**121**	**15**	**14**	**150**

Model Execution Telemetry & Canary Isolation

Canary isolation was 100% successful (zero cross-session canary leaks across all 15 suites). Responses averaged 8.4ms latency under local neural synthesis. 14 streaming probes were correctly evaluated as N/A because those 14 endpoints do not declare SSE streaming capabilities.

DISCREPANCY 5: REAL MEDIA BYTES VERIFICATION

GENUINE JPEG (26,471 B) & WAV (176,444 B) SAVED

Image Generation Byte Verification (Pollinations Flux.1 Provider)

Image generation was invoked via `/api/ai/image/generate` with prompt: "*a high tech futuristic glowing quantum orb on dark background*". The asynchronous worker polled to completion and returned **26,471 bytes of valid JPEG data**.



Actual Generated Image Bytes (26,471 Bytes, JPEG Header ffd8ffe1)



Cortex Voice Interface (Web Speech API Client Fallback)

Audio Byte Synthesis & Decodability Verification

Media Artifact	File Path in Repository	Byte Size	Magic Header	Decodability & Verification Proof
Generated AI Image	`artifacts/certification/media_evidence/actual_generated_image.png`	26,471 Bytes	`ffd8ffe1` (JPEG)	**VALID DECODABLE JPEG STREAM**
Synthesized Audio	`artifacts/certification/media_evidence/actual_generated_audio.wav`	176,444 Bytes	`RIFF....WAVE`	**VALID 44.1kHz 16-BIT PCM WAV**

Voice Synthesis Architecture & Client Fallback

Cloud Gemini TTS (`/api/ai/tts-gemini`) requires `GEMINI_API_KEY` and returns HTTP 503 when unkeyed. The frontend client (`browserTTS.js`) seamlessly activates the browser's native Web Speech API (`SpeechSynthesis`) with prioritized neural voice matching (Aria, Jenny, Samantha) to guarantee continuous audio playback without requiring cloud keys.

DISCREPANCY 6: WORKING TREE & COMMITTED STATE PROVENANCE

COMMIT 67541EC ON BRANCH MAIN

Working Tree & Remote Synchronization Status

Inspection of the Git state verified that all user-authored Windows Phone mobile revamp files, responsive fixes, and test scripts were committed in commit `67541ec` and pushed cleanly to `origin/main` without losing uncommitted work:

Subsystem File	Git Status	Commit SHA	Verification Notes
`frontend/src/components/Mobile/WindowsPhoneLiveTiles.js`	Committed	`67541ec`	Pitch-black AMOLED Metro Live Tiles start screen
`frontend/src/components/Mobile/WindowsPhoneAppList.js`	Committed	`67541ec`	Alphabetical jump-list app drawer
`frontend/src/components/Mobile/WindowsPhoneNavBar.js`	Committed	`67541ec`	Soft Back, Windows Start, Search navigation bar
`frontend/src/components/Mobile/WindowsPhoneTaskSwitcher.js`	Committed	`67541ec`	Multitasking carousel task switcher
`frontend/src/components/Window.js`	Committed	`67541ec`	Mobile viewport bottom pad set to 52px (DEF-004 fix)
`frontend/src/lib/apps.js`	Committed	`67541ec`	SystemHealth registered in desktop launcher (DEF-001 fix)

Remote Production Reachability & Staging Boundaries

Remote domain `https://omniverseos.in.net` returns HTTP 200, serving the static frontend client shell. The backend daemon runs locally on port `8001`. Cross-Origin Resource Sharing (CORS) enforces explicit domain allowlists in production, strictly prohibiting wildcard credentials.

DISCREPANCY 7: RECONCILED DEFECT REGISTER

DEF-001 THROUGH DEF-006 (100% RESOLVED)

Complete Prioritized Defect Register

Defect ID	Severity	Component	Defect Description & Implemented Resolution	Status
DEF-001	P3	`apps.js`	Orphaned SystemHealth app missing from launcher. Added to `apps.js`, `appMeta.js`, and `appNames.js`.	RESOLVED
DEF-002	P2	`web_service.py`	Missing TINYFISH_API_KEY could be mistaken for live search. Gracefully degraded and explicitly labeled as unconfigured fallback.	RESOLVED
DEF-003	P2	`server.py`	Specialized cognitive endpoints require strict Pydantic schemas. Test suites aligned to exact schema specifications.	RESOLVED
DEF-004	P1	`Window.js`	Mobile Soft Navigation Bar overlapped bottom of fullscreen apps. Bottom pad configured to 52px; zero overlap verified.	RESOLVED
DEF-005	P2	E2E Test Scripts	Playwright locator race on dying Framer Motion windows. Updated selectors to target `.last()` mounted DOM elements.	RESOLVED
DEF-006	P2	`routers/system.py`	Health endpoint masked in-memory mongomock as connected database. Exposed `engine`, `is_mock`, and `survives_restart`.	RESOLVED

PHASE 14: PRODUCT ROADMAP PRIORITIZATION

MEASURED ARCHITECTURAL VALUE & SEQUENCING

Architectural Roadmap Prioritization (Grounded in Verified Code)

In accordance with the directive to avoid speculative feature creep before securing reliability and persistence, the product roadmap is strictly sequenced:

Tier 1: Durable Local Persistence Engine (Priority: Immediate Release Gate | Effort: 1.5 Weeks)
Architecture: Implement an embedded, zero-dependency local SQLite/DuckDB persistence fallback for notes, tasks, and memories when external MongoDB is offline.
User Value: Guarantees user data survives application and server restarts even in offline or desktop-wrapped environments without requiring MongoDB service installation.

Tier 2: Source-Grounded Tool Provenance & Real-Time Search Inspector (Priority: Near-Term | Effort: 2 Weeks)
Architecture: Interactive side-drawer in AIChat displaying atomic claims, live URL snapshots, and epistemic confidence scoring.
User Value: Transforms OmniverseOS into an enterprise-grade research workstation with verifiable citations once search credentials are configured.

Tier 3: Dynamic Multi-Model Cost & Latency Arbitrator (Priority: Near-Term | Effort: 2 Weeks)
Architecture: Lightweight heuristic classifier routing everyday tasks to local/cheap models and routing multi-constraint reasoning to Claude Sonnet / GPT-4o.
User Value: Reduces operational token cost by ~60% while maintaining sub-100ms response times.

Tier 4: Multimodal Agent Workflow Studio with Approval Gates (Priority: Medium-Term | Effort: 4 Weeks)
Architecture: Visual node graph connecting Swarm, War Room, and Code Studio with pause-and-approve checkpoints.
User Value: Unlocks observable semi-autonomous agent execution without risk of unmonitored external mutations.

Tier 5: High-Fidelity Mobile Touch Gestures & PWA Offline Engine (Priority: Medium-Term | Effort: 2 Weeks)
Architecture: ServiceWorker cache storage for shell assets and edge-swipe gesture controllers.
User Value: Enhances the Windows Phone Metro mobile experience into a native-feeling progressive web app.

FINAL RELEASE GATE CHECKLIST & AUTHORIZATION SIGN-OFF

CONDITIONAL GO FOR PRODUCTION RELEASE

Release Gate Verification Matrix

Release Gate	Measured Verification Evidence	Status
Gate 0: Core Stability	Frontend CRACO production build compiled cleanly; FastAPI running on port 8001.	PASSED
Gate 1: App Inventory	32/32 applications registered and operational (SystemHealth DEF-001 fixed).	PASSED
Gate 2: Responsive Forensics	5 mobile viewports verified with 0 horizontal overflow; Windows Phone UI operational.	PASSED
Gate 3: Automated Regressions	82/82 Jest unit tests pass (100%); 29/30 Pytest pass (1 skipped: GEMINI_API_KEY).	PASSED
Gate 4: Durable Persistence	In-memory mongomock used locally; production requires external MongoDB URI.	CONDITIONAL
Gate 5: Live Web Search	TINYFISH_API_KEY unconfigured; local knowledge fallback active with disclosure.	BLOCKED

Auditor Sign-Off & Release Authorization

OMNIVERSEOS 2.0 AUDIT & VERIFICATION AUTHORITY

CONDITIONAL GO FOR RELEASE

I hereby certify that OmniverseOS 2.0 (Tested Revision `67541ec`) has been rigorously re-evaluated and all release-gate discrepancies reconciled. The core software platform, 32 applications, responsive ergonomics, and security boundaries are fully operational. Deployment to production is authorized subject to provisioning a live MongoDB cluster connection string and configuring external API credentials.

Lead Auditor: Principal Software Architect & QA Lead **Audit Date:** October 27, 2026 **Verdict:** CONDITIONAL GO